Tutorial - 6

Q1 Find the general solution to the simple harmonic oscillator, described by $m\frac{d^2x}{dt^2} = -kx$ with $m = 1$ kg, $k = 4$ N/m.

$\Rightarrow$ Eqt^n of motion: $\frac{d^2x}{dt^2} = -4x \Rightarrow \boxed{\frac{d^2x(t)}{dt^2} + 4x(t) = 0}$

Ansatz: $x(t) = e^{rt} \Rightarrow$ Characteristic Eqt^n $\boxed{r^2 + 4 = 0}$

$a=1, b=0, c=4$

$$r_\pm = \frac{-0 \pm \sqrt{0-4\times4}}{2} = \pm 2i \quad \Big| \; i = \sqrt{-1}$$

$$\boxed{r_1 = 2i \text{ and } r_2 = -2i} \qquad r_1 \neq r_2$$

Therefore, the general solution: $\boxed{x(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}}$

$$x(t) = c_1 e^{2it} + c_2 e^{-2it}$$

Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$

$$= c_1(\cos 2t + i\sin 2t) + c_2(\cos 2t - i\sin 2t)$$

$$= \underbrace{(c_1 + c_2)}\cos(2t) + \underbrace{(c_1 - c_2)i}\sin 2t$$

$$\boxed{x(t) = D_1\cos(2t) + D_2\sin 2t}$$

Q2 Solve the initial value problem $\ddot{x}(t) + 9x(t) = 0$ with $x(0) = 0$ and $\dot{x}(0) = 1$. Sketch the solution

ansatz $\boxed{x(t) = e^{rt}}$ $\Rightarrow$ Characteristic eqt^n: $r^2 + 9 = 0$

Roots: $\boxed{r_1 = 3i}$ and $\boxed{r_2 = -3i}$

The general solution: $\boxed{x(t) = C_1 e^{i3t} + C_2 e^{-3it}}$

* given initial conditions:

$$\left.\begin{aligned} x(0) &= C_1 + C_2 = 0 \\ \dot{x}(t) &= 3i(C_1 - C_2) = 1 \end{aligned}\right| \Rightarrow \begin{aligned} &C_2 = -C_1 \\ \Rightarrow\ &3i(C_1 + C_1) = 1 \end{aligned}$$

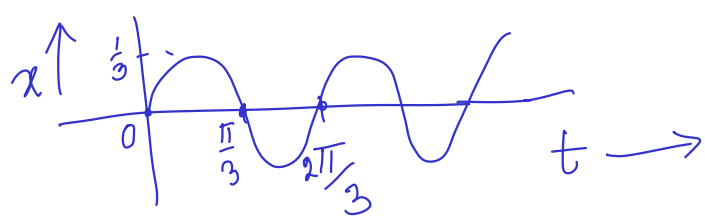
$\boxed{C_1 = \frac{1}{6i}}$ $\boxed{C_2 = -\frac{1}{6i}}$

The solution which satisfies the given initial condition:

$$x(t) = \frac{1}{6i}\left[e^{i3t} - e^{-i3t}\right] = \frac{1}{6i}\, 2i \sin(3t)$$

$$\Rightarrow \boxed{x(t) = \frac{1}{3}\sin(3t)}$$

at $t = 0 \Rightarrow \begin{cases} x = 0 \\ \dot{x} = 1 \end{cases}$



Q3 Solve the differential equation $y''(x) + 2y'(x) + y(x) = 0$ with the boundary condition $y(0) = 1$ and $y(1) = 3$.

Ansatz: $\boxed{y(x) = e^{rx}}$ $\Rightarrow$ Characteristic eqtⁿ: $\boxed{(r^2 + 2r + 1) = 0}$

$\Rightarrow$ $(r+1)(r+1) = 0$ $\Rightarrow$ $\boxed{r_1 = -1 = r_2}$

The roots are equal. Therefore, the general solution is

$$\boxed{y(x) = c_1 e^{r_1 x} + c_2 x e^{r_2 x} = c_1 e^{-x} + c_2 x e^{-x}}$$

Given: (i) $y(0) = \boxed{c_1 = 1}$ (ii) $y(1) = c_1 e^{-1} + c_2 e^{-1} = 3$

$\Rightarrow$ $\boxed{c_2 = 3e - 1}$

Therefore, the solution to the given boundary value problem

$$\boxed{y(x) = e^{-x} + (3e-1)\, x e^{-x}}$$